

Shivayokeshwari Athappan

GenAI Engineer — Large Language Models · RAG Systems · LLM-Integrated Backends

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EDUCATION

Clark University — School of Computer Science

M.S. in Computer Science | Backend Systems, Distributed Computing, Machine Learning

Dec 2025

Alagappa University

B.Sc. in Information Technology

Worcester, MA

Apr 2023

Tamil Nadu, India

EXPERIENCE

Connex AI

Oct 2025 – Present

Junior Software Developer Intern, Core Engine Team

- Engineered **3+ GenAI workflows using Python, LangChain, and OpenAI APIs** for document summarization, question generation, and context-aware processing, improving AI feature throughput by **15%**.
- Implemented **Retrieval-Augmented Generation (RAG) pipelines** over structured knowledge sources, improving contextual relevance and reducing unsupported LLM outputs across **100+ test queries**.
- Integrated **FastAPI REST services** with LLM inference, prompt templates, and Pydantic validation, reducing average backend response latency by **20%**.
- Strengthened production reliability through **PostgreSQL integration, pytest automation, structured logging, and exception handling**, increasing automated workflow coverage across **20+ backend components**.

IdeaEdu

Apr 2023 – Jun 2024

Full Stack Developer

- Delivered a **Python/FastAPI learning platform** with RESTful backend services supporting **40+ concurrent users** and end-to-end course workflows.
- Created **8+ REST API endpoints** with input validation, exception handling, and optimized request processing, reducing backend response time by **40%**.
- Designed **PostgreSQL schemas and analytics queries** for student and course data, enabling instructors to track performance across **40+ learners**.
- Launched **React.js and Chart.js dashboards** that improved student engagement by **35%** and course completion by **20%**.

PROJECTS

AI Auto Agent — RAG Document Q&A System | *Python, LangChain, RAG, FAISS, FastAPI, Streamlit* [GitHub](#)

- Architected a **RAG pipeline using LangChain, FAISS, and embeddings** to index and retrieve context from **10,000+ document chunks**, enabling grounded LLM responses.
- Optimized semantic retrieval and prompt-context construction to achieve **sub-2-second query latency** with **90%+ retrieval accuracy** across evaluation queries.
- Integrated **FastAPI** with a **3-panel Streamlit conversational interface**, enabling real-time natural-language search and document Q&A across large knowledge collections.

CleverHire — AI Resume Screener | *FastAPI, React.js, OpenAI API, PostgreSQL* [GitHub](#)

- Engineered an **LLM-powered resume screening workflow** using the OpenAI API to extract skills, evaluate job alignment, and reduce manual screening effort by **60%**.
- Designed **prompt-driven candidate scoring and gap analysis** across **100+ stored resumes**, improving candidate-to-job matching accuracy by **18%**.
- Integrated **FastAPI, PostgreSQL, React.js, and Chart.js** into an end-to-end GenAI application, while CI/CD automation reduced deployment time by **50%**.

AI-Summariser — Meeting Intelligence Assistant (MS Capstone) | *Python, FastAPI, LLMs, JavaScript* [GitHub](#)

- Created an **LLM-powered meeting intelligence pipeline** that converts transcripts into summaries, key points, decisions, action items, and calendar suggestions in a single workflow.
- Implemented a **context-aware Q&A assistant** over historical meeting transcripts, enabling natural-language retrieval across **multiple meeting records**.
- Orchestrated **FastAPI-based GenAI services** for transcript ingestion, prompt execution, structured LLM outputs, and REST integration, reducing manual post-meeting review effort by approximately **50%**.

TECHNICAL SKILLS

AI / GenAI: Generative AI (GenAI), Large Language Models (LLMs), LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Vector Databases, FAISS, Embeddings, Prompt Engineering, Natural Language Processing (NLP), Hugging Face Transformers, Scikit-learn, TensorFlow

LLM Platforms & MLOps: OpenAI API, Azure OpenAI Service, Google Vertex AI, LangSmith (LLM Tracing & Evaluation), Model Fine-Tuning (PEFT/LoRA), MLOps

Languages: Python, SQL

Backend & APIs: FastAPI, RESTful APIs, Microservices

Cloud & DevOps: AWS (EC2, RDS, S3, ElastiCache), Docker, Kubernetes, GitHub Actions CI/CD

Data & Caching: PostgreSQL, Redis